DELTA-INVARIANT LOWER BOUND VERSUS BRANCH COUNT, WITH EQUALITY GIVING THE COMPLETED MULTICROSS NORMAL FORM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the branch identification, the inequality, the completion reduction with invariance, the strict henselianness, the wedge-of-DVRs structure, and the k[[t]] normalization are all retained verbatim.

Editorial note. The label lemma-multicross-algebra names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Branches over separably closed residue fields. More on Algebra, Definition more-algebra-definition-number-of-branches. Identifies the number of branches with the number of geometric branches when the residue field is separably closed.
- The delta-branches inequality and its sharp case. Varieties, Lemmas varieties-lemma-delta-number-branches-inequality and varieties-lemma-delta-number-branches-inequality-sh. Supplies the general inequality and, in the strictly henselian equality case, the wedge-of-DVRs structure.
- Invariance under completion and the coefficient field. More on Algebra, Lemma more-algebra-lemma-one-dimensional-number-of-branches; Varieties, Lemma varieties-lemma-delta-same-after-completion; Algebra, Lemma algebra-lemma-complete-henselian and algebra-lemma-regular-complete-containing-coefficient-field. Lets one pass to the completion without changing the invariants, get strict henselianness, and write each complete DVR branch as k[[t]].

Source reference. Original lemma 04GM: sources/stacks-chow/tags.txt, source lines 5838–5838. Prerequisite pointer invoked at proof line(s) 3454; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C0S: sources/stacks-chow/tags.txt, source lines 15097–15097. Prerequisite pointer invoked at proof line(s) 3458; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0C1U: sources/stacks-curves/pinned-original-source.tex, source lines 3410–3410. Prerequisite pointer invoked at proof line(s)

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3437; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0C26: sources/stacks-chow/tags.txt, source lines 15146–15146. Prerequisite pointer invoked at proof line(s) 3444; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C2D: sources/stacks-chow/tags.txt, source lines 15153–15153. Prerequisite pointer invoked at proof line(s) 3451; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C43: sources/stacks-chow/tags.txt, source lines 15213–15213. Prerequisite pointer invoked at proof line(s) 3446; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C42: sources/stacks-chow/tags.txt, source lines 15212–15212. Prerequisite pointer invoked at proof line(s) 3455; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C3W: sources/stacks-chow/tags.txt, source lines 15206–15206. Prerequisite pointer invoked at proof line(s) 3452; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a separably closed field. Let A be a 1-dimensional reduced Nagata local k-algebra with residue field k. Then*

$$\delta\text{-invariant } A \geq \text{number of branches of } A - 1$$

If equality holds, then $A^\wedge$ is as in (0C1U).

Proof. Since the residue field of A is separably closed, the number of branches of A is equal to the number of geometric branches of A, see More on Algebra, Definition 0C26. The inequality holds by Varieties, Lemma 0C43. Assume equality holds. We may replace A by the completion of A; this does not change the number of branches or the δ-invariant, see More on Algebra, Lemma 0C2D and Varieties, Lemma 0C3W. Then A is strictly henselian, see Algebra, Lemma 04GM. By Varieties, Lemma 0C42 we see that A is a wedge of complete discrete valuation rings. Each of these is isomorphic to $k[[t]]$ by Algebra, Lemma 0C0S. Hence A is as in (0C1U). □